$$v_c'(\text{at}) = -\frac{v_0}{2}$$

b、c meet at equal speeds leftward: b、c do not collide again. At b、c's second meeting, (17)、(18), (1) give the four displacements from their own equilibria.

$$\left.\begin{aligned} x_a' &= 0 \\ x_b' &= 0 \\ x_c' &= 0 \\ x_d' &= 0 \end{aligned}\right\}$$

Both springs are natural. The four table velocities are, respectively,

$$\left.\begin{aligned} v_a &= \frac{v_0}{2} + \frac{v_0}{2} = v_0 \\ v_b &= \frac{v_0}{2} - \frac{v_0}{2} = 0 \\ v_c &= \frac{v_0}{2} - \frac{v_0}{2} = 0 \\ v_d &= \frac{v_0}{2} + \frac{v_0}{2} = v_0 \end{aligned}\right\} \tag{19}$$

a right, b still: compression speeds up b, slows a. c rests, d moves: stretch speeds up c, slows d. The spring oscillators resume oscillating.

From (16), when

$$\omega_2 t = 2\pi$$

$$\omega_1 t = 4\pi$$

b、c meet again: b、c center-frame velocities by (2) are

$$v_b'(\text{at}) = -\frac{v_0}{2}$$

$$v_c'(\text{at}) = \frac{v_0}{2}$$

b left, c right, equally fast: b、c meet without impact. The four table velocities are

$$\left.\begin{aligned} v_a &= v_0 \\ v_b &= 0 \\ v_c &= v_0 \\ v_d &= 0 \end{aligned}\right\} \tag{20}$$

Then a、b mass center M_1 moves right uniformly at $\frac{1}{2}v_0$.

a、b move harmonically about M_1 at angular frequency ω_1. c、d have mass center M_2. It moves uniformly right at $\frac{1}{2}v_0$. c、d do so about M_2 at angular frequency ω_2. b、c meet every $\frac{\pi}{\omega_2}$ time units, always in either case above: meeting without colliding.